Model to Predict the Probability of a Criminal Event in Peru Using Machine Learning Algorithms

2025 8th International Conference on Artificial Intelligence and Big Data (ICAIBD) | 979-8-3315-1936-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICAIBD64986.2025.11082025

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***Abstract*—Insecurity in the Callao (province of Peru) region has intensified in recent years, despite the presence of strategically distributed police stations. This issue is influenced by a lack of integration of advanced technologies in crime prevention, limiting the effectiveness of security strategies. To address this situation, the present study proposes a predictive model based on Machine Learning algorithms applied to the crime context of Callao-Peru. The methodology includes three main stages: Analysis of relevant features, model design, and results validation. In the analysis, eight variables (F1 to F8) associated with criminal patterns were evaluated, while the validation phase compared the performance of the algorithms in terms of accuracy (ACC), precision, F1-score, recall, and AUC. The results show that the Decision Tree algorithm had the highest overall accuracy, achieving an ACC of 0.97, followed by Gradient Boosting (ACC = 0.93) and Random Forest (ACC = 0.92), while K-Nearest Neighbors had the lowest performance (ACC = 0.72). These findings position the Decision Tree as the most effective tool for predicting high-risk crime areas in Callao-Peru.**

Keywords—machine learning, criminality, crime prediction, decision tree

I. Introduction

The police stations in the Callao region, despite being geographically well-distributed, have not managed to reduce insecurity among residents due to the lack of a comprehensive approach to crime prevention [1]. On the other hand, in Rio de Janeiro, urban violence has had a considerable adverse effect on the local economy, particularly in tourism, where a significant decrease in tourists has been observed due to the high rate of violent crimes [2]. Similarly, there is a lack of analysis to develop better technological solutions that could leverage human mobility patterns from GPS trajectory data, providing greater security along routes [3].

In Callao, insecurity remains one of the main concerns of residents, with 7,516 complaints filed in the second quarter of 2023, heightening the sense of vulnerability [4]. However, over the past year, INEI [5] recorded 4,345 complaints in Callao between July and August 2023. These crimes are primarily related to offenses against property (62.6%), public safety (13.1%), and life, body, and health (5.6%).

To mitigate citizen insecurity, various studies have explored advanced technological approaches. Muñoz [6] developed Machine Learning (ML) models to predict crimes in Medellín, establishing a comparative baseline of algorithms. Héctor Dimitri [7] proposed a collaborative mobile system in Bogotá to identify unsafe areas, while Sueldo Alarcón and Peña Cueva [8] created a web service to predict and prevent thefts in Metropolitan Lima using artificial intelligence. Additionally, Sossa Jhonatan [9] introduced a methodology to anticipate crimes through a binary classification system, although it has yet to be applied in daily practice.

However, research focused on the prediction and identification of crimes in the specific context of Callao remains scarce. A review of the literature indicates that most of these studies have centered on other geographic realities. Therefore, this article proposes a predictive model based on ML designed to identify high-crime-risk areas in Callao. By leveraging historical and geospatial data, the model aims to detect criminal patterns and key factors associated with risk, contributing to greater effectiveness in crime prevention strategies in the region.

The document is organized as follows. Related works are discussed in Section 2. The details of the proposed system architecture are presented in Section 3. Section 4 covers the results and discussion protocol. Finally, the conclusions are presented in Section 5.

II. Related Works

In the literature, various studies have been developed related to different types of criminal events, applying advanced technologies and various ML algorithms.

A. Types of Criminal Events

Studies focus on various types of crimes that significantly impact urban areas, such as robbery, theft, assault, rape, and homicide. In [10], the focus is on violent crimes such as 'homicide,' 'violent robbery,' and 'rape' in Mexico City, seeking patterns in the most vulnerable areas to improve prevention strategies. In [11], common crimes such as 'robbery,' 'theft,' and 'assault' are analyzed, applying ML models to predict these recurrent crimes in urban areas. The study uses historical and geographical data to identify high-risk areas. In [12], the analysis primarily covers 'theft', including 'motorcycle theft' and property loss in urban areas. This study conducts a systematic review emphasizing the use of advanced ML and Deep Learning (DL) techniques to improve crime prediction in large cities. In

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[13], the focus is also on crimes like 'robbery' and 'assault,' using historical data to build predictive models that help authorities target their efforts in specific areas with higher crime incidence. In [15], crimes such as 'homicide' and 'armed robbery' are included, applying DL architectures to predict and classify these crimes in urban contexts. The study's focus is on improving prediction accuracy using large volumes of unstructured data, such as images and texts. Finally, in [16], the use of IoT technology for detecting and monitoring suspicious activities in real-time is highlighted; although it does not focus on a specific type of crime, its application has a significant impact on preventing 'robberies' and 'assaults' through constant urban area monitoring.

B. Technologies

Studies apply various technologies such as ML (Machine Learning), DL (Deep Learning), and the Internet of Things (IoT) to improve crime prediction and prevention. In [14], the use of 'IoT' is explored for the detection and tracking of criminals in urban environments, enabling real-time monitoring of suspicious activities. This approach enhances the response capability of security forces and optimizes resources allocated for surveillance. Meanwhile, in [16], these systems are integrated into smart city infrastructures, facilitating the collection and analysis of large volumes of real-time data, which allows for the early identification of potential threats.

In [11], 'ML' is used to predict crimes in San Francisco through algorithms such as Random Forest (RF) and Gradient Boosting Decision Trees (GBDT), allowing authorities to identify high-crime areas and make proactive decisions. On the other hand, [12] conducts a systematic review highlighting the use of algorithms such as RF and XGBoost to predict criminal patterns globally by integrating geographic and demographic data. In [13], ML is applied to crime classification using approaches such as K-Nearest Neighbors (KNN) and Decision Trees (DT), helping authorities focus their resources on areas with higher crime incidence. Finally, in [15], models combining ML with smart cities are explored, demonstrating how the integration of both technologies enhances cities' capacity to anticipate and prevent crimes.

In [12], 'DL' is used to develop predictive crime models through architectures such as Convolutional Neural Networks (CNNs), enabling the processing of large volumes of complex data, such as images and texts, thereby improving the accuracy of crime event classification. In [14], it is used for detecting anomalous patterns in smart cities, helping to identify unusual behaviors from multiple data sources. Meanwhile, in [15], DL is applied to both crime classification and prediction, standing out for surpassing traditional ML approaches in handling large amounts of unstructured data.

C. Machine Learning Algorithms

Various ML algorithms have been used in studies to improve the accuracy of crime prediction (see Table I). In [12], 'RF' (Random Forest) is used to predict crimes in smart cities, highlighting its ability to handle complex spatial and temporal data, making it suitable for crime prevention in urban areas. In [11], the same algorithm is applied in San Francisco to create a predictive model that helps authorities enhance surveillance in specific areas. In [11], 'Naive Bayes (NB)' focuses on its speed in processing large volumes of data, making it useful for rapid classification applications, although less precise than more complex models. In [13], 'KNN' (K-Nearest Neighbors) is used to analyze the geographic proximity of crimes, allowing the identification of patterns in high-crime areas and facilitating targeted preventive efforts. In [11], 'GBDT' (Gradient Boosting Decision Trees) stands out as an effective tool for refining predictions through an iterative approach that corrects errors, helping to identify hidden patterns and improving data-driven decisions. Finally, in [13], 'DT' (Decision Trees) are valued for their simplicity and speed, enabling immediate operational decisions, which is useful in scenarios where speed is more important than precision.

TABLE I. MACHINE LEARNING ALGORITHMS

Algorithm	ACC -AUC	Ref.	Variables
RF	ACC: 97%	[12]	Historical crime data, points of interest, demographic, and weather data
	ACC (Training): 80.84% ACC (Testing): 63.43%	[11]	Date, type of crime, crime resolution, geographic location, time of day
NV	ACC (Training): 65.82% ACC (Testing): 64.33%	[11]	Date, type of crime, geographic location, crime resolution, day of the week
KNN	ACC 75.89%	[13]	Date, type of crime, geographic location, crime resolution, police district
GBDT	ACC (Training): 99.44% ACC (Testing): 99.75%	[11]	Date, type of crime, police district, crime resolution, day of the week, geographic coordinates
DT	ACC 83.6%	[13]	Date, type of crime, geographic location, crime resolution, police district

III. PROPOSED MODEL

This section presents the proposed model for analyzing crime data in Callao. Key phases of the process are described, such as data selection and preprocessing, as well as the implementation of ML algorithms used to identify criminal patterns and predict high-risk areas.

A. Dataset Selection

For this study, data obtained from the National Open Data Platform of Peru, specifically related to crime in the Callao region, were used [17]. The choice of this dataset is based on the fact that Callao has the highest crime rates in Peru, making it an ideal context for the application of predictive crime models.

The dataset contains detailed information on various aspects of more than 2,000 crimes recorded in February 2024, comprised of 26 attributes, all documented in Spanish, such as: fecha_corte, fecha_registro, id_doc_denuncia, ubigeo, departamento, provincia, distrito, tipo_de_denuncia, situacion_denuncia, tipo, subtipo, modalidad, fecha_hecho, hora_hecho, ubicacion, descripcion, fecha_nacimiento, edad_persona, sexo, estado_civil, grado_instruccion, ocupacion,

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pais_natal, mes, longitud, and latitud. These variables provide a comprehensive overview of the geospatial and temporal characteristics of each incident.

B. Data Preprocessing

Data preprocessing is an essential phase to ensure that the dataset is in the proper condition to be used by ML algorithms. The data obtained from the National Open Data Platform was processed through several key steps. Initially, null values, duplicate records, and inconsistent data were removed, prioritizing entries with complete geospatial and temporal information. Temporal variables such as the date and time of crimes were standardized, and geographic coordinates were aligned to ensure consistency. The most relevant crimes for the analysis (homicide (C1), theft (C2), robbery (C3), and rape (C4)) were selected, and categorical variables were numerically encoded. Balancing techniques, such as oversampling and undersampling, were applied to address class imbalance, preventing biases toward more frequent crimes and improving the accuracy of the model.

During this phase, eight features (F1 to F8) were identified as crucial for predicting crime patterns. These features were selected based on exploratory data analysis and their relevance within the dataset attributes. F1 and F2 represent the geographic coordinates (latitude and longitude) of the crime location, capturing its spatial distribution. F3 corresponds to the crime modality or classification, reflecting the nature of the incident. Temporal features such as F4 (month), F5 (time of day: morning, afternoon, or evening), and F6 (day of the week) were included to analyze seasonal and daily crime patterns. F7 represents the distance from the crime location to the nearest police station, providing insight into the relationship between law enforcement presence and crime incidence. Finally, F8 indicates the district where the crime occurred, offering a territorial perspective within the Callao region. These features were carefully selected to ensure that the dataset captures geospatial, temporal, and situational factors critical to crime prediction. Table II shows the main characteristics of the dataset that were later used for training.

TABLE II. DATASET CHARACTERISTICS

ID	Feature	Description
F1	Latitud_denuncia	Geographic coordinate indicating the north-south position where the report was filed
F2	Longitud_denuncia	Geographic coordinate indicating the east-west position of the location where the crime occurred
F3	Modalidad	Classification of the type of crime or the manner in which the criminal act was carried out
F4	Mes	Indicates the month of the year when the crime occurred
F5	turno	Time of day when the crime occurred. This attribute is derived from the hora_hecho field in the original dataset
F6	Dia_semana	Day of the week when the report was filed. This attribute is derived from the fecha_hecho field in the original dataset
F7	Distancia_comisaria	Distance in meters from the crime location to the nearest police station
F8	distrito	Territorial division within the jurisdiction of Callao where the report was filed
F9	Sub_tipo	Classification of the type of crime that occurred at the specific location and time. This is what will be predicted

C. Algorithm Training

Figure 1 illustrates the training flow steps related to the prediction of 'Criminal Events' using the RF model in Python. The process begins with training the model using the train_test_split function, which handles data division into a test set and a training set, creating the variables X_train, X_test, y_train, y y_test to allocate 20% of the data to the test set and the remaining 80% to the training set (Step 1). The RF algorithm is called using sklearn's ensemble library and the model variable is created to manage training and predictions. The model is fitted using the X_train and y_train variables via the fit function, allowing the model to learn from the training data (Step 2). Predictions are made with the trained model. The variable test_pred contains predictions made using the X_test data. Then, the confusion_matrix function is used to create a confusion matrix, stored in the conf_matrix variable, which shows the correct and incorrect predictions for each output class. Additionally, the model's accuracy is calculated using accuracy_score, with the accuracy stored in test_acc (Step 3). A detailed classification report is generated using the classification_report function. This report provides key evaluation metrics for the model, such as precision, recall, and F1-score for each criminal class (crime subtypes), showing the model's performance for each prediction category (Step 4 Finally, the flow concludes by displaying the model's results and validating its performance based on the test set data.

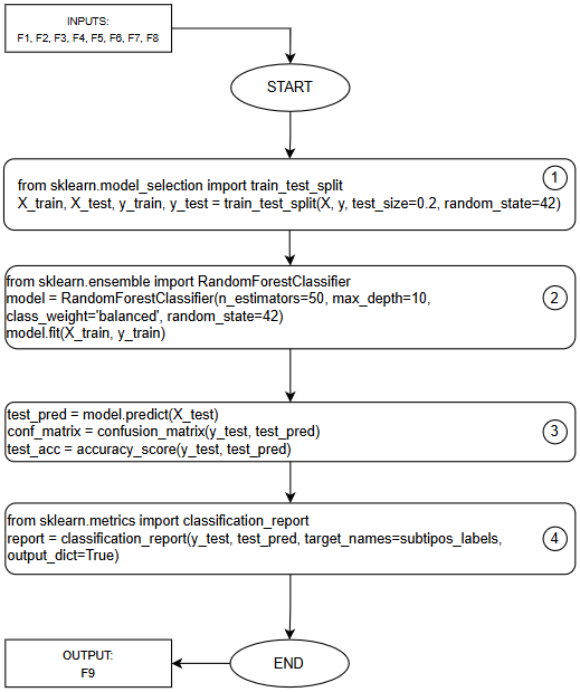


Fig. 1. ML Model Training Flow

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This process is used for all algorithms. Additionally, the same process is followed for training related to crime prediction.

D. Evaluation

The model evaluation was conducted using standard metrics to ensure its performance in terms such as: Accuracy (Eq. 1), Precision (Eq. 2), Recall (Eq. 3), and F1-Score (Eq. 4).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Where:

- TP (True Positives): Correctly classified case of a crime type.
- TN (True Negatives): Correctly classified case of a non-crime type.
- FP (False Positives): Incorrectly classified case of a crime type.
- FN (False Negatives): Incorrectly classified case of a non-crime type.

To evaluate the model's performance, several key metrics were employed. Accuracy measures the overall proportion of correct predictions, while Precision focuses on the quality of positive predictions, minimizing false positives. Recall assesses the model's ability to correctly identify crimes (minimizing false negatives). Additionally, the F1-Score combines Precision and Recall, providing a balanced measure in contexts where classes are imbalanced. Lastly, the AUC measures the model's ability to distinguish between different classes, with a value close to 1 serving as a key indicator of good performance in class discrimination.

IV. Results and Discussion

To evaluate the effectiveness of the models, performance metrics were calculated using confusion matrices (Fig. 2), which allowed for the identification of the algorithms that achieved the best results in terms of correct and incorrect predictions for crime prediction.

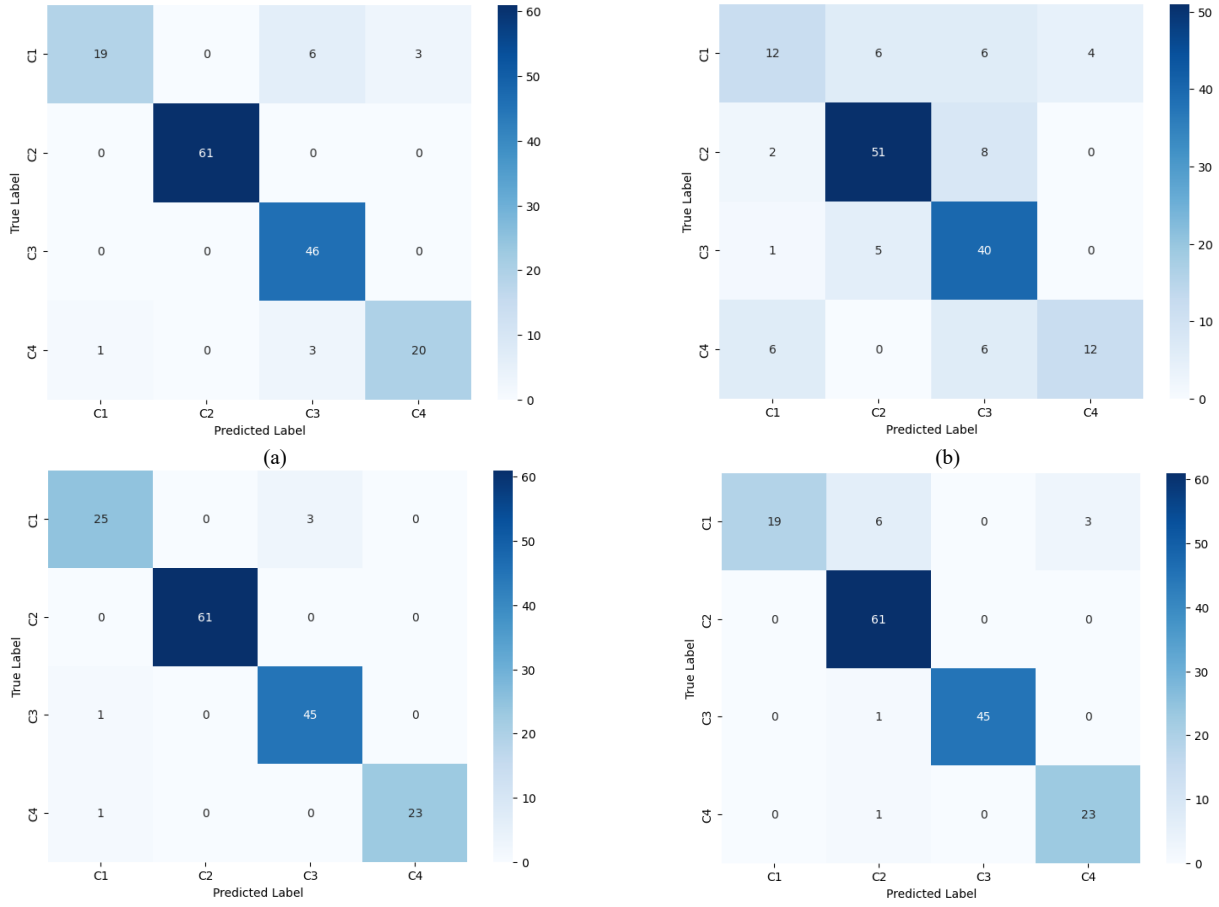


Fig. 2. Confusion matrices for predictions made by (a) RF, (b) KNN, (c) DT, (d) GBDT

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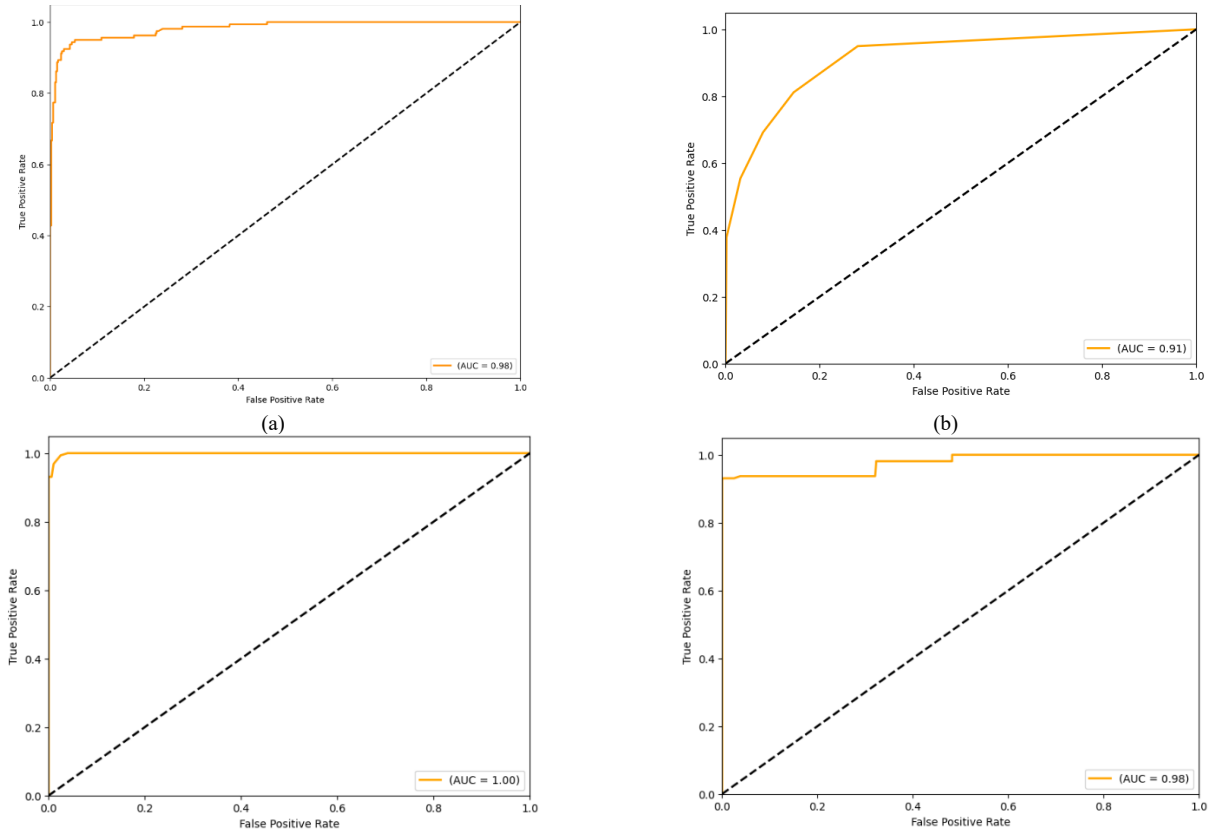


Fig. 3. ROC curves of the algorithms used in model development (a) RF, (b) KNN, (c) DT, (d) GBDT

Table III shows the number of correct and incorrect results per class and algorithm. In this regard, the model using the DT algorithm achieved the highest number of correct predictions with 154, followed by GBDT, RF, and KNN with 148, 146, and 115 correct predictions, respectively.

Fig. 3 shows the comparison of the ROC curve results for the four algorithms. It is observed that DT achieved the best performance with an AUC of 1.00, indicating perfect classification. It is followed by RF and GBDT, both with an AUC of 0.98, suggesting excellent performance with few incorrect predictions. KNN, with an AUC of 0.91, has a lower performance in comparison, generating more incorrect predictions. Overall, all algorithms perform well, with DT being the most accurate.

TABLE III. CORRECT AND INCORRECT PREDICTIONS OF CRIMINAL EVENTS

Algorithm	Class	Correct prediction	Incorrect prediction
RF	C1	19	9
	C2	61	1
	C3	46	0
	C4	20	4
KNN	C1	12	16
	C2	51	10
	C3	40	6
	C4	12	6
DT	C1	25	3
	C2	61	0
	C3	45	0
	C4	23	1
GBDT	C1	19	9
	C2	61	0
	C3	45	1
	C4	23	0

Table IV shows the comparison of algorithms with respect to the previously defined metrics. It is evident that the DT algorithm performs best, achieving an Accuracy of 0.97, followed by GBDT with 0.93 and RF with 0.92. Regarding crime prediction, the criminal event 'theft' (C2) achieved the highest precision with the RF algorithm. The criminal events 'homicide' (C1) and 'robbery' (C3) had the best precision with the GBDT algorithm. The criminal events 'theft' (C2) and 'rape' (C4) had the highest precision with the DT algorithm. Additionally, the criminal event 'theft' (C2) achieved the best precision with KNN.

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TABLE IV. COMPARISON OF METRICS BY ALGORITHM

ML Model	Class	Precision	F1-Score	Recall	AUC	ACC
RF	C1	0.95	0.79	0.68	0.98	0.92
	C2	1.00	1.00	1.00		
	C3	0.84	0.99	0.98		
	C4	0.87	0.85	0.83		
GBDT	C1	1.00	0.81	0.68	0.98	0.93
	C2	0.88	0.94	1.00		
	C3	1.00	0.99	0.98		
	C4	0.88	0.92	0.96		
DT	C1	0.93	0.91	0.89	1.00	0.97
	C2	1.00	0.96	0.98		
	C3	0.94	1.00	1.00		
	C4	1.00	0.98	0.96		
KNN	C1	0.57	0.49	0.43	0.91	0.72
	C2	0.82	0.83	0.84		
	C3	0.67	0.75	0.87		
	C4	0.75	0.60	0.50		

V. CONCLUSION

In this study, four ML algorithms, RF, GBDT, KNN, and DT, were evaluated for crime prediction in Callao. Eight features (F1 to F8) were used to train the models and analyze their predictive capabilities in an urban security context. Through this analysis, it was observed that each algorithm has varying performance in its ability to correctly predict crimes, affecting its applicability in urban security settings.

The DT model stood out with the highest precision in classes C2 (theft) and C4 (violation), while GBDT performed best in classes C1 (homicide) and C3 (robbery). RF and KNN also showed good precision in class C2 (theft). Overall, DT was the algorithm with the highest overall precision, standing out as the most effective model for distinguishing between different types of crimes.

In terms of DT, although it achieved a perfect AUC of 1.00, indicating excellent performance on the ROC curve, its confusion matrix reveals the presence of some false negatives. While this number is small, it could have implications in situations where omitting serious crimes is unacceptable. Despite its excellent theoretical precision, this small margin of error could be critical in practical implementation, especially in contexts where reliability is essential. Nonetheless, its high AUC positions it as a valuable tool in situations where model interpretability is important, in addition to its predictive capability.

ACKNOWLEDGMENT

We would like to thank the Research Department of the Universidad Peruana de Ciencias Aplicadas for the support provided in carrying out this research work.

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